



NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA

B.Tech. I Semester (CSE/IT/AIIML/IOT/MC/MV/ADS/PI)

Mid Term – I Examination

Date: Sept. 24, 2025

Max. Marks: 15

Time: 10:00 am – 10:50 am

Paper: Differential Calculus and Differential Equations (MAIC 101)

All questions carry equal marks

Instructions: The calculator is not allowed into the examination hall, No extra answer sheets will be provided.

1. a). Find the general solution of a non-homogeneous system of linear equations using Row Echelon form:
 $x + y + z + t = 1, y + z + t = 0, x + z = 0.$
b). Also, express this general solution of the above system as the sum of two non-zero solutions, say X_1 and X_2 such that X_1 is the solution corresponding to a homogeneous system of linear equations: $x + y + z + t = 0, y + z + t = 0, x + z = 0$ and X_2 is the solution corresponding to the original non-homogeneous system of linear equations: $x + y + z + t = 1, y + z + t = 0, x + z = 0.$
2. $Q = 2xy$ is a quadratic form in three variables over $\mathbb{R}$.
a). Express Q in matrix vector form. b). Write the type of the matrix A associated with Q
c). Find the eigen values of A . d). Write the canonical form of Q .
e). Find the eigen vectors of A . f). Find the transformation which transform Q into its canonical form.
g). Find the orthogonal transformation which transform Q into its canonical form.
h). Find the rank, index and signature.
i). Is the definition of rank, as used in the above question, applicable to every square matrix?
3. Show that the function $f(x, y) = \begin{cases} \frac{xy}{\sqrt{x^2+y^2}}: (x, y) \neq (0,0) \\ 0: (x, y) = (0,0) \end{cases}$ is continuous at $(0,0)$ and $f_x(0,0)$ and $f_y(0,0)$ exists.